



The Deutsch Proposition

The research paper by David Deutsch that forms the basis for the time travel in Endgame. <https://digit.in/june19-1>



Endgame accuracy

Which of the terms in Endgame were used for effect, and which were consistent? <https://digit.in/june19-2>



THE QUANTUM PHYSICS OF ENDGAME

Avengers: Endgame does not hesitate to dive into obscure theories of how the universe works

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At the end of Avengers: Infinity War, Thanos had collected all the Infinity Stones to make the complete the Infinity Gauntlet, and then uses it to wipe half the living things in the universe out of existence. In Ant Man and the Wasp, Ant Man gets stuck in the quantum realm as the Wasp, and Hank Pym who were supposed to bring him back, are all removed from existence by the actions of Thanos. Fans had already speculated that Ant Man would be the key to the solution, using some kind of time travel. How

this actually plays out in Endgame is very different from other movies that deal with time travel. Endgame appears to get down and dirty with quantum physics, exploring some of the less well known theories in the process. If you have not watched the movie yet, and care about spoilers, the story is discussed in detail beyond this point, so it is better to come back after catching the film.

TIME TRAVEL

One of the earliest depictions of time travel was in the 1843 novel, A Christmas Carol by Charles Dickens, where Ebenezer Scrooge gets to visit Christmases in the past and future. In 1895, HG Wells popularised the notion of using devices for time travel with The Time Machine. The phrase “time machine” was coined in this novel, which is now the commonly used term for such a device. Just a few years later, Einstein formulated his theory of relativity, which had many jaw dropping implications. One of these was time dilation, where the time measured by one person

could differ from other person. If you are an observer watching a friend go by in a car, time is passing slightly slower for the person who is moving faster. If there are two identical clocks, and one of them is under a stronger gravitational influence, time will pass more slowly for that clock. Time passes at a different rate for everyone because of this reason. These differences in time dilation are most apparent at relativistic speeds, or when a clock is travelling close to the speed of light. In which case hardly any time passes by. Time dilation has been proved experimentally using pairs of incredibly accurate atomic clocks, one sent to space and the other on Earth. The Global Positioning System works because all the atomic clocks on the satellites correct the effects of time dilation to ensure that they are synced to the clocks on Earth. Astronaut Scott Kelly spend a year on board the space station, and when he came back, the age gap with his twin brother increased by 5 milliseconds. Time dilation shows that time travel is possible in one direction. If someone, let's call him Rip Van Winkle, orbits the Earth in a spacecraft travelling at nearly the speed of light allowing twenty years to pass on Earth, when they land, they have practically jumped twenty years into the future. Now the question arises, if time travel is possible into the past?

ENTER THE WORMHOLE

Kip Thorne, a Nobel prize winning physicist has been a consultant for a number of science fiction works, including Contact and Interstellar. It was Thorne who suggested Worm Holes as a scientifically plausible way of quickly traversing interstellar distances, when Carl Sagan was writing Contact. Wormholes are theoretical structures, also known as Einstein-Rosen bridges, that can connect two points in the space time continuum. A wormhole has two mouths, or ends. Now imagine one of these mouths is moving relativistically to the other, similar to the spaceship of Rip Van Winkle. This mouth then moves close to where the first mouth begins. If Rip